WAVEU3S: A LIGHTWEIGHT WAVELET DUAL-ATTENTION UNET FOR 3D MEDICAL IMAGE SEGMENTATION

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ABSTRACT

Medical image segmentation is a key step in medical image analysis. Compared with the convolutional network where the receptive field is usually limited, the transformer model is able to extract the long-range dependencies between various locations and thus suitable for this task. However, for 3D medical image segmentation, the computation cost increases dramatically with the spatial resolution. In this paper, we propose a lightweight hybrid model (WaveU3S) for 3D medical segmentation based on Unet framework and 3D wavelet transform with dual attention. With 3D wavelet transform, the input features are compressed to half spatial resolution without losing information, reducing the computation burden while maintaining the performance. Experiments on FLARE22 and ACDC datasets demonstrate that the proposed method can produce satisfied segmentation result at a low cost.

***Index Terms*—** Medical image segmentation; Vision transformer; 3D wavelet;

1. INTRODUCTION

Medical image segmentation is of vital importance in computer vision and medical imaging. It aims to classify anatomical structures or areas of interest in a medical image into different categories or labels so that a physician or computer system can analyze the anatomical structures and lesion areas of interest [1].

Convolutional neural network (CNN) is one of the most important tools in the field of medical image segmentation. It can learn complicated feature representations from enormous volumes of medical images and have done well at tasks like tumor identification, organ segmentation, and disease diagnosis [2, 3, 4]. However, CNN has limited receptive fields and is unable to capture long-range dependencies. Fortunately, with the introduction of ViT [5] into the field of medical image processing, transformer-based and CNN-transformer-based segmentation networks are proposed to deal with long-range dependencies, demonstrating outstanding performance and ability [5]. However, when it comes to segmentation tasks for 3D medical images,computational volume becomes a prominent issue. When we compute features with volume size $H \times W \times D$ and a number of C channels, the computational complexity of self-attention is typically $\mathcal{O}\left[(HWD)^2 C\right]$. Therefore, its application in medical image analysis is quite limited with restricted computational resources and time. To tackle with this problem, recent works [6, 7, 8] usually adopt either downsampling [6], linear projection [7], or local attention [8] to reduce the computation burden. Convformer [9] uses maximum pooling and convolution to reduce the spatial resolution. UNETR++ [10] and DCAMSR [11] project the input feature via a linear layer to reduce the computational amount of spatial attention. Cotr [12] minimizes the computational complexity by flexible self-attention calculation only on a small number of critical areas. Similarly, attention computation is carried out at the patch level on a non-global scale by [13, 14]. It is crucial to note that all of these techniques could result in information loss.

Inspired by Wave-vit [15] where the input feature is compressed with 2D wavelet before self attention calculation, we propose a lightweight Wavelet dual-attention Unet for 3D medical image Segmentation (WaveU3S). Unlike other methods for reducing space, 3D wavelet transform can reduce spatial resolution while keeping the detailed information in the original features in channel. Thanks to the invertibility of 3D wavelet transform, the high resolution feature can be reconstructed after the attention calculation, thus reducing the computation while maintaining the performance. Our contributions are threefold: 1) 3D wavelet decomposition module is designed for spatial resolution compression of feature maps without losing information. 2) A lightweight wavelet dual-attention model is proposed based on the 3D wavelet module to incorporate both spatial and channel attention for segmentation. 3) Experiments on multiple segmented datasets demonstrate that the proposed model can achieve satisfied results with few parameters and computation.

2. METHODOLOGY

2.1. Overall Framework

The Wave3US model utilizes multi-resolution framework, and the whole network has a U-shaped architecture, as shown in Fig. 1 (a). For encoder, Wavelet Compression Dual-

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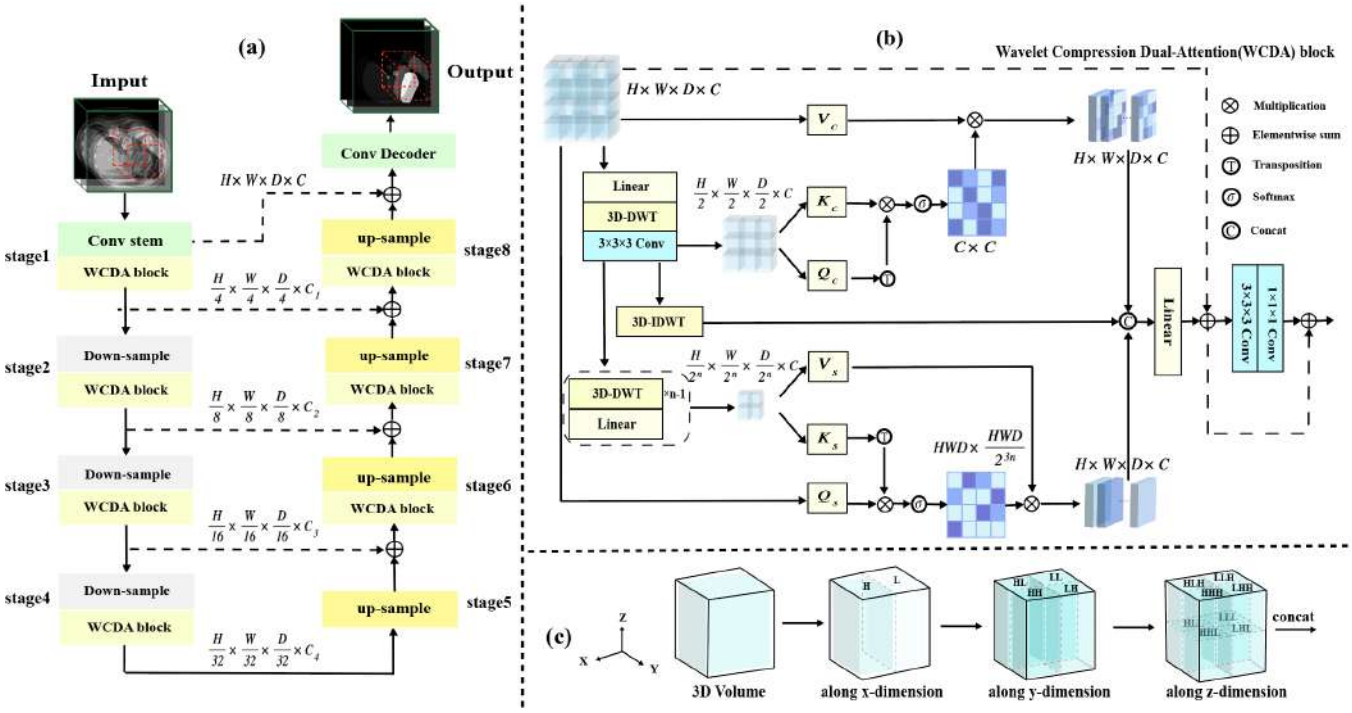


Fig. 1: Diagram illustration of the proposed method. (a) Architecture of WaveU3S. (b) Architecture of the Wavelet Compression Dual-Attention (WCDA) block. (c) Diagram of 3D wavelet decomposition.

Attention (WCDA) block is used in each stage to extract features at various feature resolutions $\chi \in \mathbb{R}^{\frac{H}{2^i} \times \frac{W}{2^i} \times \frac{D}{2^i}}$. Similarly, the decoder consists of four stages, where a WCDA block and an upsampling layer of a transposed convolution are connected to the corresponding encoding stage. The output of the last decoder is fused with the highest resolution feature map to recover spatial information and enhance the feature representation.

2.2. Wavelet Compression Dual-Attention (WCDA) Block

The WCDA block, shown in Fig 1 (b), contains several wavelet compression blocks and a dual-attention transformer. The wavelet compression block is used for the compression of the input features to achieve a lossless reduction of spatial resolution of the feature map. The dual-attention transformer is used to capture the inter-channel relationship of the features as well as the dependency between spatial locations, providing rich contextual information for the model.

2.2.1. 3D Discrete Wavelet (3D-DWT) Layer

Fig. 1 (c) demonstrates the decomposition of a 3D volume using 3D-DWT. Since wavelet transform is separable, the decomposition is conducted along the x, y, z axes sequentially to get eight sub-bands.

To further compress the feature, we employ a linear transformation to reduce the channel size of the input 3D feature map $X \in \mathbb{R}^{H \times W \times D \times C}$ by a factor of 8, resulting in $X' \in \mathbb{R}^{H \times W \times D \times \frac{C}{8}}$, where X' is the transformed feature map with reduced channel size. Then we downsample it using the 3D-DWT, yielding eight wavelet subbands: X_{LLL}, X_{LLH}, X_{LHL}, X_{LHH}, X_{HLL}, X_{HLH}, X_{HHL}, X_{HHH}, where the size of each component is $X_{\text{subband}} \in \mathbb{R}^{H/2 \times W/2 \times D/2 \times C/8}$. The calculation method is as follows:

$$X_{\alpha\beta\gamma} = \text{Conv}(X', W_{\alpha\beta\gamma}), \quad \forall \alpha, \beta, \gamma \in \{L, H\}, \tag{1}$$

$$W_{\alpha\beta\gamma} = (f_\alpha \otimes f_\beta) \otimes f_\gamma, \quad \forall \alpha, \beta, \gamma \in \{L, H\}, \tag{2}$$

where the subband components $X_{\alpha\beta\gamma}$ are computed by applying the convolution operation between the input feature map X' and the wavelet filter $W_{\alpha\beta\gamma}$. Each filter $W_{\alpha\beta\gamma}$ is constructed through a tensor product of filters along three dimensions, where $\alpha, \beta, \gamma \in \{L, H\}$. For simplicity but without loss of generality, we use Harr wavelets in the study. The eight wavelet sub-bands are then concatenated along the channel dimension, yielding the compressed feature map $X_{dwt} \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2} \times C}$.

2.2.2. Dual-attention transformer

The input features and their compressed versions are then used to calculate dual attention. Specifically, we project the original input into Q_s and V_c, and the output of the compressed features after one 3D-DWT, denoted as $X^1_{\text{dwt}} \in$

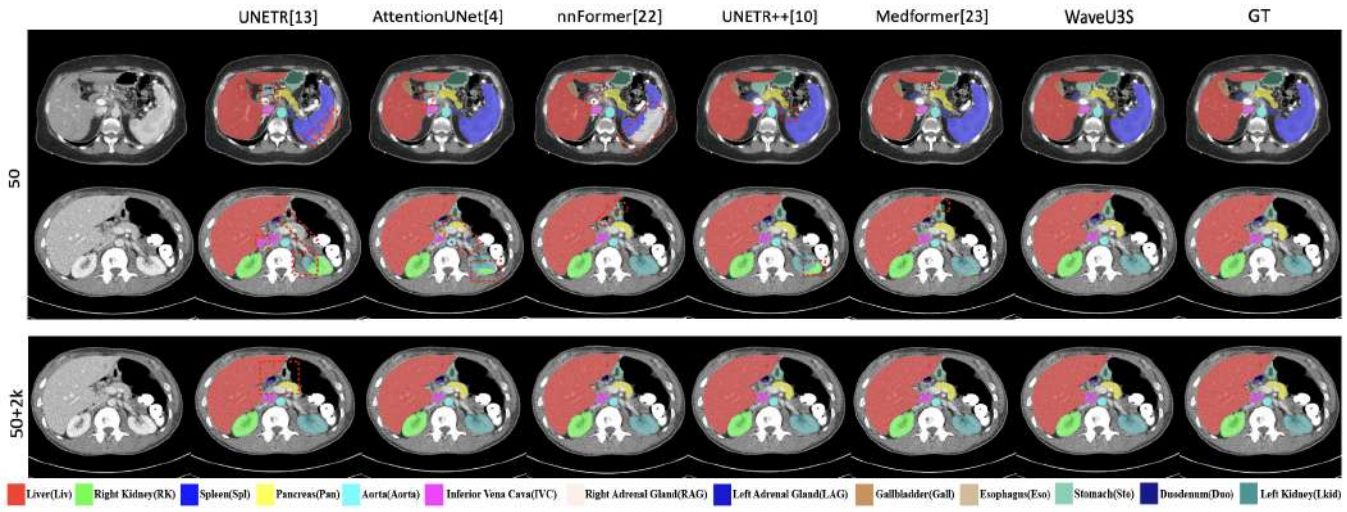


Fig. 2: Qualitative comparison on the FLARE22 dataset. The text within the parentheses in the figure represents the abbreviations for various organs.

Table 1: Quantitative comparison on the FLARE22 dataset, where the best result is in **bold** and the second best result is underlined. Results training on labeled scans alone and mixed with unlabeled data are denoted as 50 and 50+2k, respectively.

Training	Algorithm	Liv.	RK.	Spl.	Pan.	Aorta.	IVC.	RAG.	LAG.	Gall.	Eso.	Sto.	Duo.	LKid.	Avg.
50	UNETR [13]	90.12	73.11	77.46	62.21	90.76	78.17	67.32	62.06	52.87	72.90	67.28	51.93	71.10	70.56
	AttentionUNet [4]	96.08	83.95	93.14	80.04	95.45	83.81	74.76	70.08	63.35	81.12	85.82	67.39	88.81	81.18
	nnFormer [16]	96.94	81.72	90.73	81.27	96.60	85.05	**83.00**	80.53	58.29	82.49	84.24	69.96	85.02	82.76
	UNETR++ [10]	**97.37**	81.24	93.37	85.12	**97.15**	**88.17**	82.63	78.83	66.88	79.74	84.99	66.68	**95.52**	84.43
	MedFormer [17]	97.00	**89.80**	94.05	83.14	94.55	87.09	78.56	74.53	**72.14**	**85.48**	**87.86**	**71.91**	86.25	84.80
	WaveU3S(ours)	97.35	86.02	**96.90**	**85.47**	96.89	86.88	79.58	**80.57**	65.17	81.28	84.34	69.34	94.46	**84.94**
50+2k	UNETR [13]	96.95	87.27	93.49	81.48	95.63	87.49	78.88	73.03	64.55	81.11	85.88	66.62	87.61	83.07
	AttentionUNet [4]	98.00	92.40	97.13	88.22	96.72	89.05	82.29	80.47	80.95	87.01	91.19	80.61	89.48	88.73
	nnFormer [16]	97.68	90.97	95.10	87.45	96.58	89.16	79.28	78.08	75.36	83.73	91.49	75.63	90.42	87.00
	UNETR++ [10]	98.06	**95.66**	**97.62**	90.56	97.03	90.84	84.36	81.90	80.16	88.72	92.06	82.28	**91.63**	90.07
	MedFormer [17]	**98.30**	95.40	97.58	90.79	**97.18**	**91.38**	**85.45**	**84.37**	**82.66**	**89.37**	92.44	82.46	90.29	**90.59**
	WaveU3S(ours)	98.14	93.32	97.41	**91.02**	96.94	90.88	84.69	82.05	82.35	88.86	**92.93**	**82.71**	90.34	90.09

$\mathbb{R}^{\frac{H}{2}\times\frac{W}{2}\times\frac{D}{2}\times C}$, are projected into K_{c} and Q_{c}. The features compressed by $n-1$ 3D-DWTs, denoted as $X_{\mathrm{dwt}}^{n} \in \mathbb{R}^{\frac{H}{2^n}\times\frac{W}{2^n}\times\frac{D}{2^n}\times C}$, are projected into V_{s} and K_{s}. Followings are the calculations for spatial attention and channel attention.

$$\hat{X}_s = \mathrm{Softmax}\left(\frac{Q_s K_s^T}{\sqrt{d}}\right) V_s, \tag{3}$$

$$\hat{X}_c = \mathrm{Softmax}\left(\frac{Q_c^T K_c}{\sqrt{d}}\right) V_c, \tag{4}$$

where Q_{s} and Q_{c} represent the query matrices for spatial and channel attention. K_{s} and K_{c} represent the key matrices for spatial and channel attention. V_{s} and V_{c} represent the value matrices for spatial and channel attention. The scaling factor d is learnable per head, adjusting attention dynamically .

The complexity of channel attention and spatial attention obtained by our method is $O(C^2\frac{HWD}{2^3})$ and $O(\frac{H^2W^2D^2}{2^{3n}}C)$, respectively. Similar to WaveVit, we use a companion 3D inverse wavelet transform layer for restoring the original spatial information and details X_{idwt} for fusion with subsequent attention features.

$$X = \mathrm{Linear}\left(\mathrm{Concat}\left(\hat{X}_s, \hat{X}_c, X_{\mathrm{idwt}}\right)\right) \tag{5}$$

We then pass the fused result through a convolutional stem to get the final output of WCDA block.

2.3. Loss Function

Dice and cross-entropy losses are used to train the model.

3. EXPERIMENTAL AND RESULTS

3.1. Datasets

The FLARE22 [18] is a CT dataset, assembled with permission under license from over 20 medical groups, including but not limited to MSD [19], KiTS [20] and AbdomenCT-1K

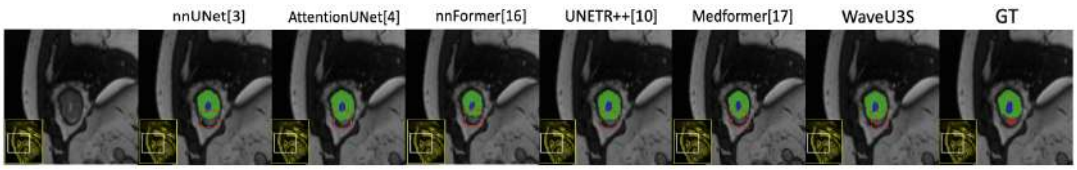


Fig. 3: Qualititive comparison on the ACDC dataset.

Table 2: Comparison of parameters and GFLOP, with input size of [96,192,192], where the best result is in **bold** and the second best result is underlined.

Algorithm	Params(M)	GFLOPs
UNETR [13]	92.60	330.82
AttentionUNet [4]	**16.60**	559.75
nnFormer [16]	149.67	1215.28
UNETR++ [10]	44.50	**233.10**
Medformer [17]	32.31	1892.28
WaveU3S (ours)	24.70	233.23

Table 3: Quantitative comparison on the ACDC dataset, where the best result is in **bold** and the second best result is underlined.

Algorithm	RV	Myo	LV	Avg
nnUNet [3]	91.02	90.15	93.95	91.71
AttentionUNet [4]	**92.53**	90.89	94.41	92.61
nnFormer [16]	90.18	88.73	93.31	90.74
UNETR++ [10]	92.34	91.09	94.49	92.79
Medformer [17]	91.02	89.61	92.52	91.05
WaveU3S (ours)	92.41	**91.23**	**95.02**	**92.89**

[21]. The training set comprises 50 labelled and 2000 unlabeled scans. The model in Huang et al.[22] is employed to generate pseudo-labels for the 2000 unlabeled scans to serve as additional pseudo-labeled training data. The test set consists of 50 cases with label.

The ACDC dataset [23] an MRI dataset for segmentation of the left ventricular (LV), right ventricular (RV) and myocardial (Myo) structures. It includes 100 training samples and 50 testing samples.

The input 3D volume is of 96×192×192 for the FLARE22 dataset and 12×192×192 for the ACDC dataset, respectively. For both datasets, we divide the training set into training and validation sets with a ratio of 4:1. Dice Similarity Coefficient (DSC) on the test set is chosen as the evaluation metric.

3.2. Implementation Details

For fair comparision, we use nnUNet [3] as default data preprocessing method and training framework, with the same train/validation partition. The comparison state-of-the-art methods include UNETR [13], AttentionUNet [4], nnFormer [16], UNETR++ [10] and MedFormer [17]. All models are trained with random initial weights, with PyTorch 1.11.0, Python 3.9.7 and CUDA 11.3 on a single NVIDIA RTX 3090, except for MedFormer which is trained on an NVIDIA RTX A6000 due to its large GPU memory consumption.

3.3. FlARE22 Result

The quantitative results are shown in Table 1. Trained only on the 50 labelled samples, the proposed method is already superior to UNETR and AttentionUNet; and comparable to nnFormer, UNETR++ and MedFormer, with a slightly higher average DSC. When adding unlabelled trainning data, the proposed method can also obtain satisfied performance, with the best result for 3 categories and the second best result for 6 categories as well as the average. Fig. 2 shows the visual results. It can be seen that our network is able to distinguish between right kidney and left kidney, compared with other methods. Besides, as shown in Table 2 WaveU3S has the second least parameters and GLOPs among all these methods: compared with UNETR++, WaveU3S has comparable GFLOPs but with 44% fewer parameters; compared with AttentionUNet, Wave3US has about half more parameters but with only 44% GLOPs.

3.4. ACDC Result

The quantitative results are shown in Table 3, where WaveU3S gets the highest DSC for LV, Myo and the average, as well as the second best DSC for RV. As shown in Fig. 3, only MedFormer and WaveU3S can segments the LV out.

4. CONCLUSIONS

In this paper, we propose a lightweight 3D medical image segmentation network termed as WaveU3S, where the feature extraction block (WCDA) is based on 3D wavelet compression and dual-attention feature fusion. With lossless 3D wavelet compression, the WCDA bolck can calculate both the spatial and channel attention efficiently. Experiments on FLARE22 and ACDC datasets demonstrate that the proposed method can produce satisfied results with less parameters and computational complexity than existing methods.

5. HUMAN AND/OR ANIMAL SUBJECTS

This research study was conducted retrospectively using human subject data made available in open access by FLARE22 and ACDC. Ethical approval was not required as confirmed by the license attached with the open access data.

6. ACKNOWLEDGMENTS

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